Autoformalization

Melchizedek, Amrit

Programming Languages Lab
Dept. of Computer Science and Automation
Indian Institute of Science, Bangalore

Guide: Deepak D'souza

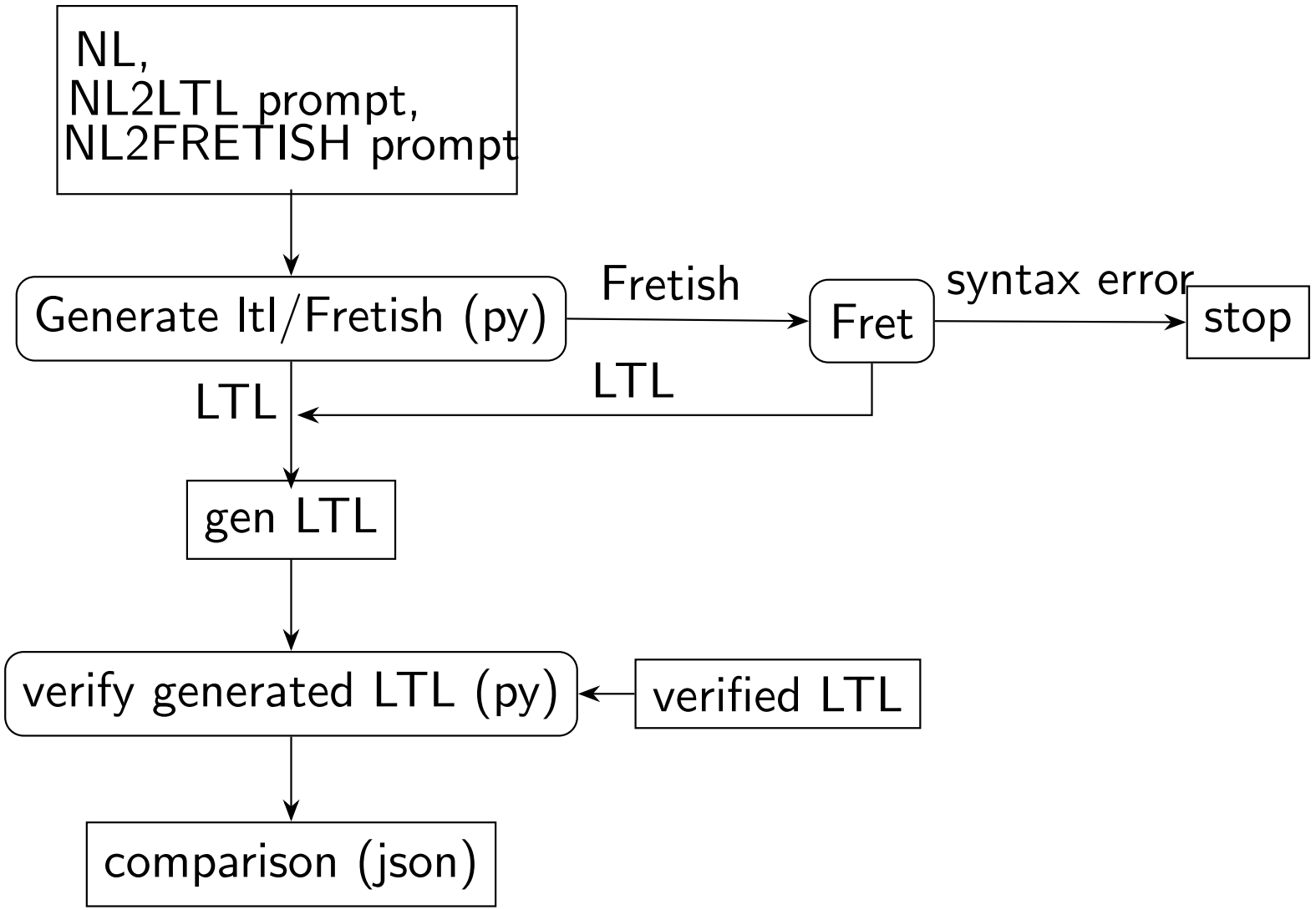
Data Set

- Data set is taken from VLTL Bench
- Contains 3 dataset: "Search and rescue", "traffic light" and "ware house".
- Randomly 1000 NL chosen
- Made 920 verified LTL formulas for the NL
- 10: Few shot example for the prompt, 910: used to benchmark

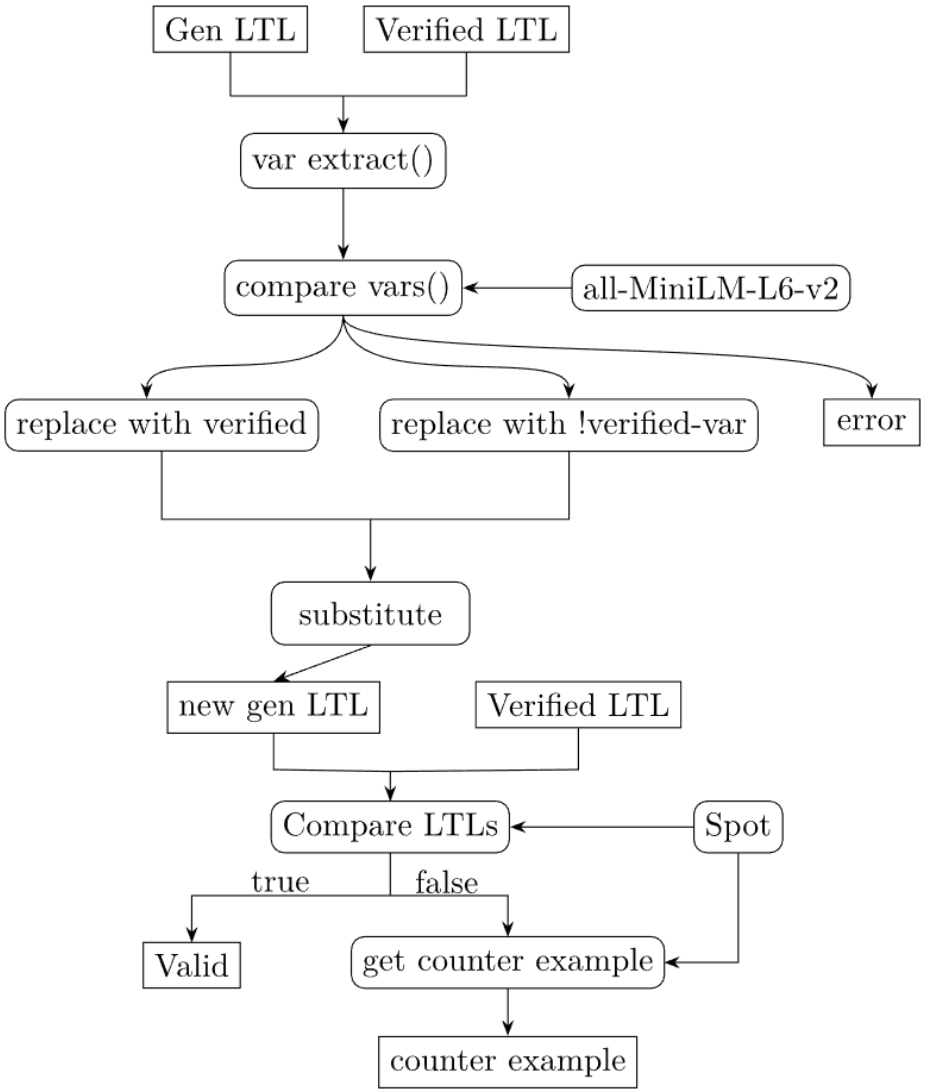
Benchmarks

- Model: Deepseek v4 flash
- Input : NLtoLTL-prompt, NLtoFRETish-prompt, NL-dataset

Work Flow


NL,
NL2LTL prompt,
NL2FRETISH prompt
Generate ltl/Fretish (py)
Fretish
Fret
syntax error
stop
LTL
LTL
gen LTL
verify generated LTL (py)
verified LTL
comparison (json)

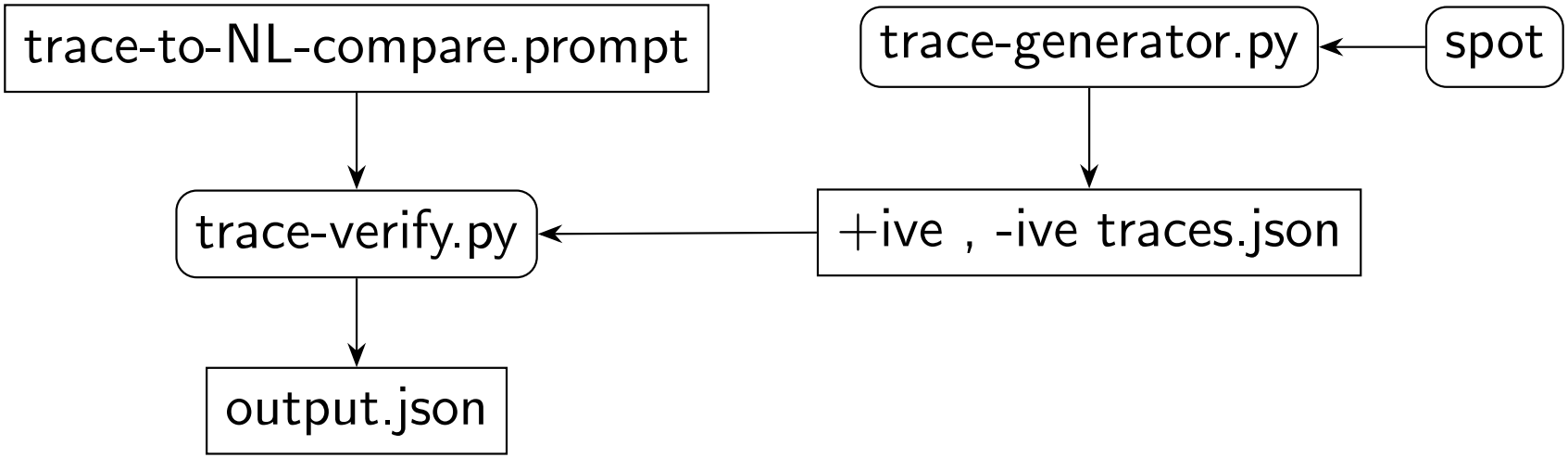
Verifying LTL



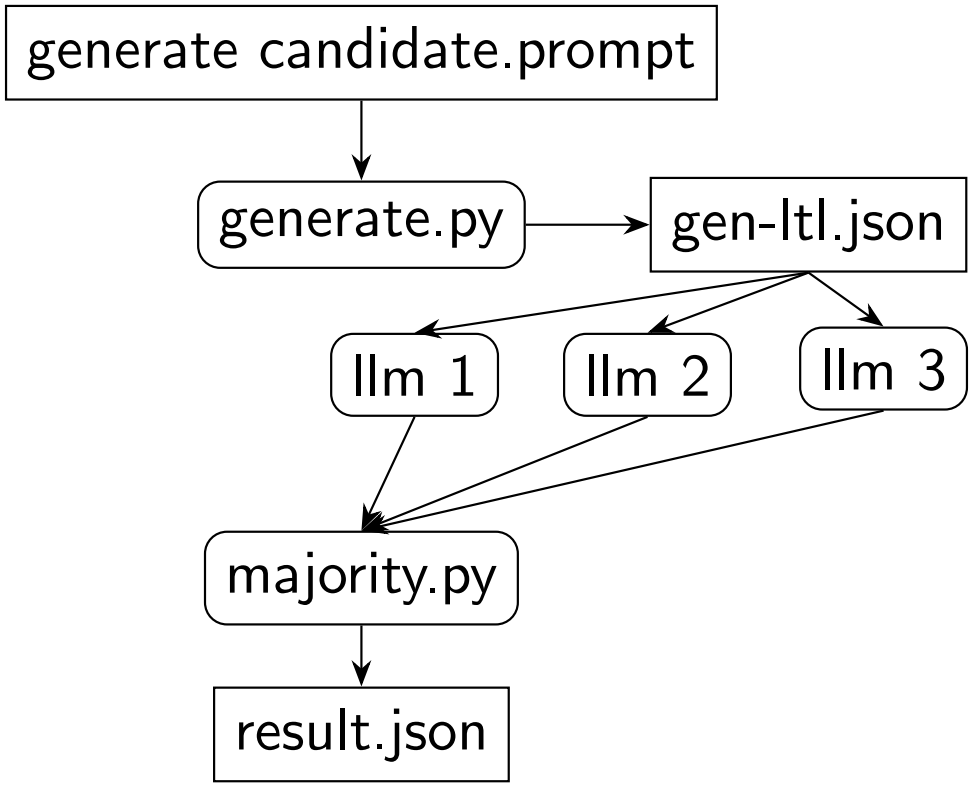
Results

- Total: 920
- True: 675
- False: 228
- errors: 7 (response contained other data)
- More results Here.

Traces


trace-to-NL-compare.prompt
trace-generator.py
spot
trace-verify.py
+ive , -ive traces.json
output.json

Majority Voting



n candidates

